

ROADSAFE
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**SMART ROAD
SURFACE DAMAGE
DETECTION SYSTEM**

GROUP 09

K.G.C.Pahasara	ICT/2023/040
D.M.M.M.Mahanama	ICT/2023/042
M.G.J.N.Wijesinghe	ICT/2023/039
M.R.Fatheena	ICT/2023/079

**ICT 2403
Graphics & Image
Processing**



INTRODUCTION

About the Project

Road conditions impact safety and vehicle performance. This project proposes a video based road damage detection system, utilizing mobile phone videos and image processing to identify and assess road damage automatically. It offers a low-cost solution suitable for developing countries, addressing the limitations of traditional manual road inspections.

Problem Statement

Current road inspection methods are primarily manual and dependent on human judgment, leading to inefficiencies such as overlooked minor damages, inaccurate severity assessments, and reliance on costly advanced tools. These limitations adversely affect road safety and maintenance planning.

Proposed Solution

The proposed system automates road damage detection using video data.

- Road videos are captured using a mobile phone
- Videos are divided into frames for analysis
- Image processing techniques are applied to detect damaged areas
- Common damages detected:
 - Potholes
 - Alligator cracks
 - Edge Cracks
 - Lane Removings

Damage severity is calculated using size, area, and shape, and classified as follows and a colorcoded road severity map is generated:

- Minor: Green — Low severity
- Moderate: Yellow Medium severity
- Severe: Red — High severity

The system also compares past and current data to predict future damage growth.

Technology Used

- Video Processing: Mobile phone video capture and frame extraction
- Image Processing: Grayscale conversion, noise removal, edge detection
- Severity Analysis: Rule-based classification
- Software: Python and OpenCV

Outcome and User Benefits

The system provides outputs such as the current road damage severity, a road damage risk score, and a prediction of future road conditions. These results help save time and inspection costs, reduce human involvement, enable early damage detection, improve maintenance planning, enhance road safety, and offer an effective solution for low-budget environments